

PRACTICAL – 10

Part – A

Output:

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-10$ echo "This is the original file." > original.txt
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-10$ ls -li original.txt
18155 -rw-r--r-- 1 bhardwaj bhardwaj 27 Oct  1 09:24 original.txt
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-10$ |
```

```
18155 -rw-r--r-- 1 bhardwaj bhardwaj 27 Oct  1 09:24 original.txt
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-10$ stat original.txt
  File: original.txt
  Size: 27                Blocks: 8          IO Block: 4096   regular file
Device: 8,48      Inode: 18155        Links: 1
Access: (0644/-rw-r--r--)  Uid: ( 1000/bhardwaj)   Gid: ( 1000/bhardwaj)
Access: 2026-10-01 09:24:32.656755878 +0000
Modify: 2026-10-01 09:24:37.932330008 +0000
Change: 2026-10-01 09:24:37.932330008 +0000
 Birth: 2026-10-01 09:24:32.656755878 +0000
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-10$ |
```

```
18155 -rw-r--r-- 1 bhardwaj bhardwaj 27 Oct  1 09:24 original.txt
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-10$ ln original.txt hardlink.txt
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-10$ ls -li original.txt hardlink.txt
18155 -rw-r--r-- 2 bhardwaj bhardwaj 27 Oct  1 09:24 hardlink.txt
18155 -rw-r--r-- 2 bhardwaj bhardwaj 27 Oct  1 09:24 original.txt
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-10$ |
```

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-10$ ls -li original.txt hardlink.txt symlink.txt
18155 -rw-r--r-- 2 bhardwaj bhardwaj 27 Oct  1 09:24 hardlink.txt
18155 -rw-r--r-- 2 bhardwaj bhardwaj 27 Oct  1 09:24 original.txt
18180 lrwxrwxrwx 1 bhardwaj bhardwaj 12 Oct  1 09:26 symlink.txt -> original.txt
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-10$ |
```

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-10$ find . -inum $(ls -li original.txt | awk '{print $1}')
./original.txt
./hardlink.txt
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-10$ |
```

```
./hardlink.txt
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-10$ rm original.txt
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-10$ ls -li hardlink.txt
18155 -rw-r--r-- 1 bhardwaj bhardwaj 27 Oct  1 09:24 hardlink.txt
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-10$ cat hardlink.txt
This is the original file.
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-10$ |
```

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-10$ ls -l symlink.txt
lrwxrwxrwx 1 bhardwaj bhardwaj 12 Oct  1 09:26 symlink.txt -> original.txt
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-10$ cat symlink.txt
cat: symlink.txt: No such file or directory
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-10$ |
```

Part-B

Code:

```
GNU nano 7.2 mmap_file.c
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>
#include <sys/mman.h>
#include <sys/stat.h>
#include <string.h>

int main()
{
    const char *filename = "mmap_data.txt";
    const char *text = "Hello from memory mapped file!\n";

    int fd = open(filename, O_RDWR | O_CREAT, 0666);

    if (fd == -1)
    {
        perror("open");
        return 1;
    }

    size_t size = strlen(text);

    if (ftruncate(fd, size) == -1)
    {
        perror("ftruncate");
    }
}
```

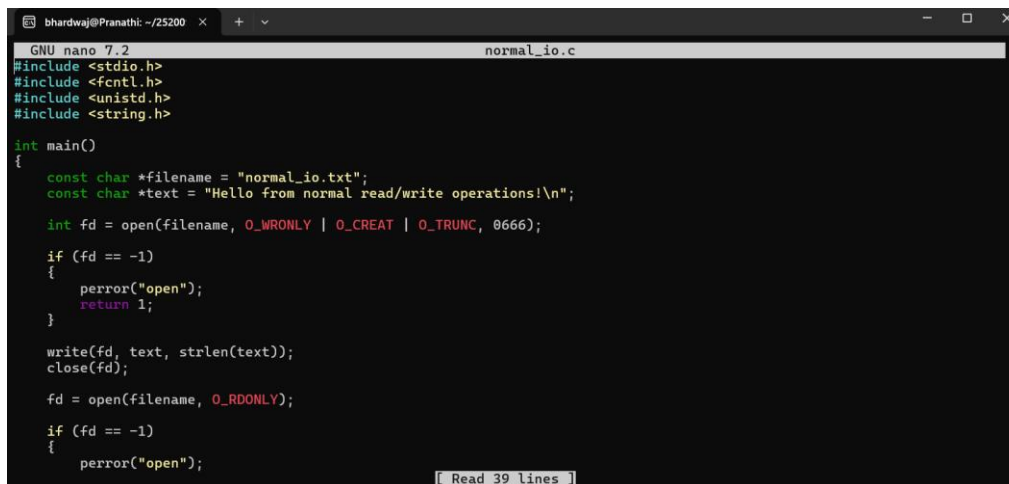
Output:

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-10$ ./mmap_file
Data written using mmap(): Hello from memory mapped file!
File successfully written using memory mapping.
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-10$ |

bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-10$ cat mmap_data.txt
Hello from memory mapped file!
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-10$ |
```

Part C:

Code:



```
GNU nano 7.2 normal_io.c
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <string.h>

int main()
{
    const char *filename = "normal_io.txt";
    const char *text = "Hello from normal read/write operations!\n";

    int fd = open(filename, O_WRONLY | O_CREAT | O_TRUNC, 0666);

    if (fd == -1)
    {
        perror("open");
        return 1;
    }

    write(fd, text, strlen(text));
    close(fd);

    fd = open(filename, O_RDONLY);

    if (fd == -1)
    {
        perror("open");
    }
}
```

Output:

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-10$ nano normal_io.c
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-10$ gcc normal_io.c -o normal_io
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-10$ ./normal_io
Data read using normal read(): Hello from normal read/write operations!
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-10$ |

bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-10$ ls -lh mmap_data.txt normal_io.txt
-rw-r--r-- 1 bhardwaj bhardwaj 31 Oct  1 09:29 mmap_data.txt
-rw-r--r-- 1 bhardwaj bhardwaj 41 Oct  1 09:31 normal_io.txt
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-10$ cat mmap_data.txt
Hello from memory mapped file!
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-10$ cat normal_io.txt
Hello from normal read/write operations!
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-10$ |
```